Graph Based Manga Recommendation System

CSE 694 – Deep Learning with Graphs

Project Checkpoint

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Project Recap

1

Team: Myself

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Project: Manga Recommender

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Goal: Extend existing content based recommender using graph methods

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Core: Model users, manga, and metadata as graphs

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Idea: Treat recommendation as a link prediction problem

Motivation

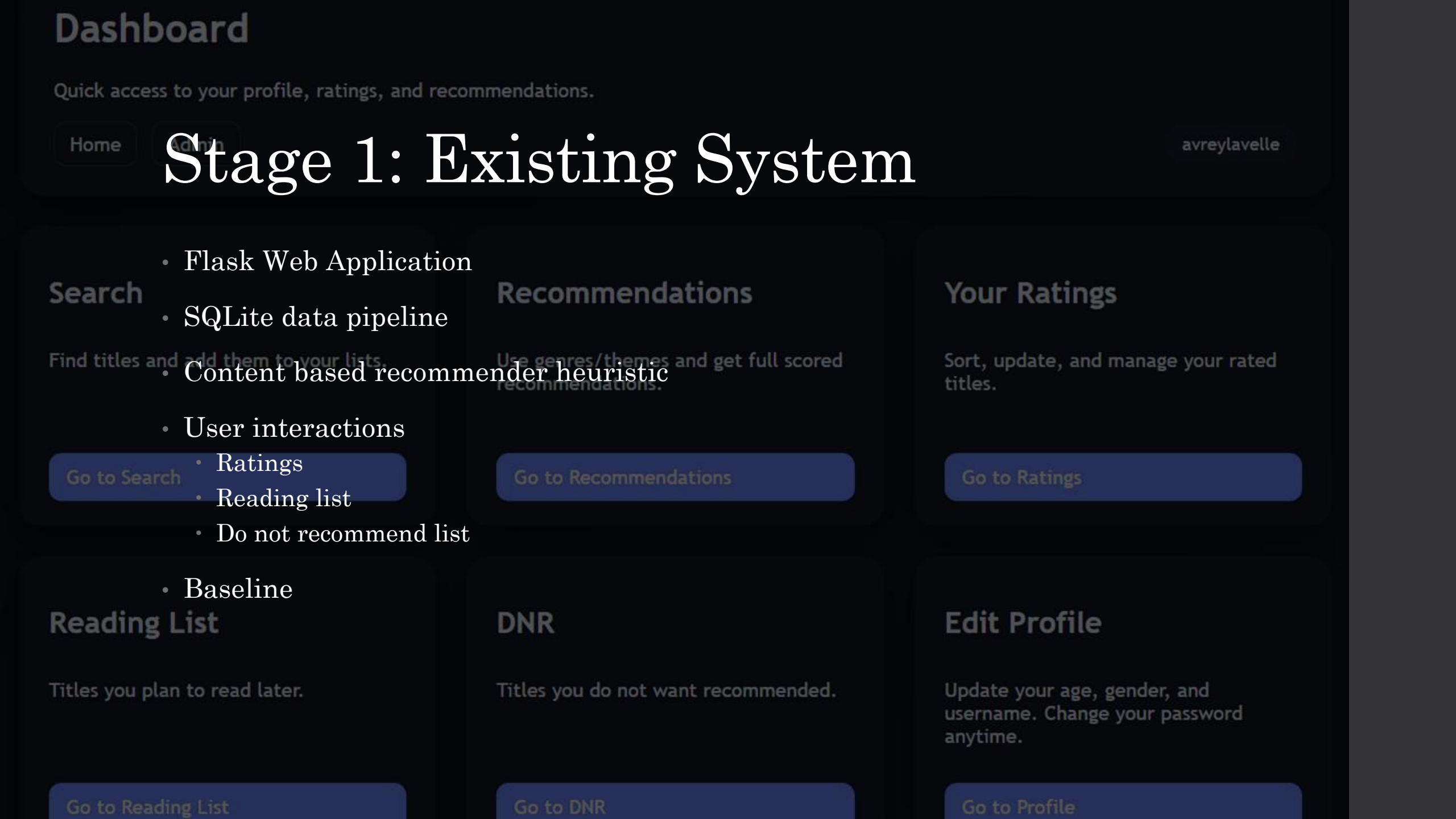
- Traditional recommender systems rely on content similarity and popularity rankings
- This ignores relationships between users, mangas, genres, themes, authors, etc.
- Graph models allow us to represent and exploit those relationships.

Dataset

- SQLite Dataset constructed from
- MyAnimeList user data
- Mangadex metadata
- Manga.db

Stage 1: Existing System

- Flask Web Application
- SQLite data pipeline
- Content based recommender heuristic
- User interactions
  - Ratings
  - Reading list
  - Do not recommend list
- Baseline



Stage 2 Objective

Methods implemented:

Common Neighbors
Jaccard Similarity
Adamaic-Adar Index

Baselines used:

V3 Recommender
Popularity Ranking (MyAnimeList)

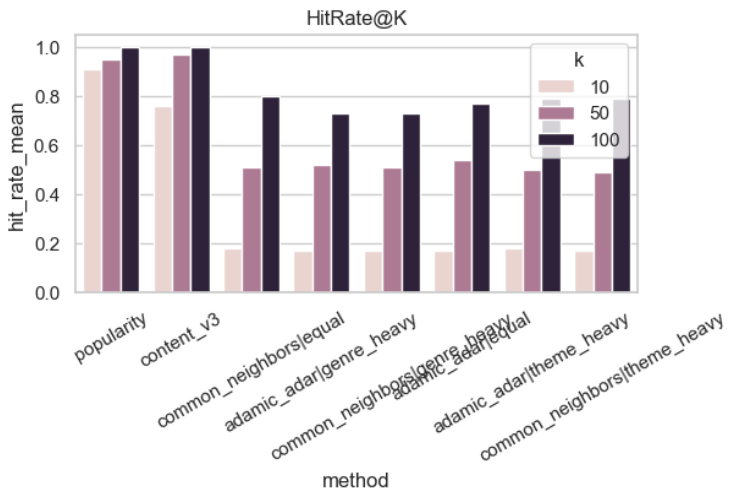
Evaluation Metrics:

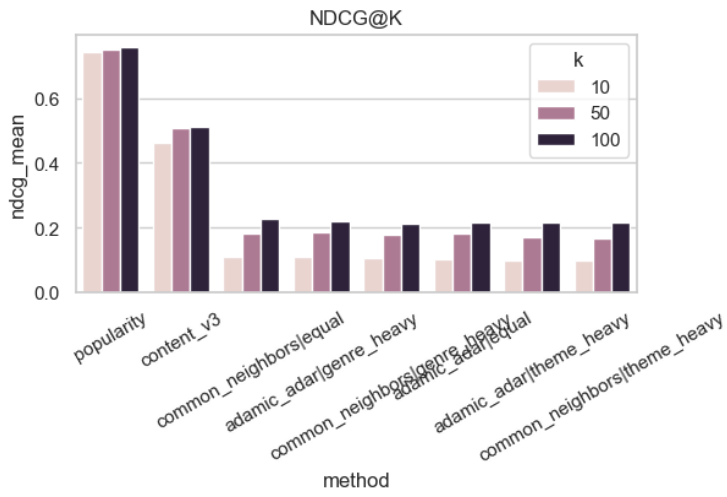
Precision@K
Recall@K
NDCG@K
Hitrate@K
MRR

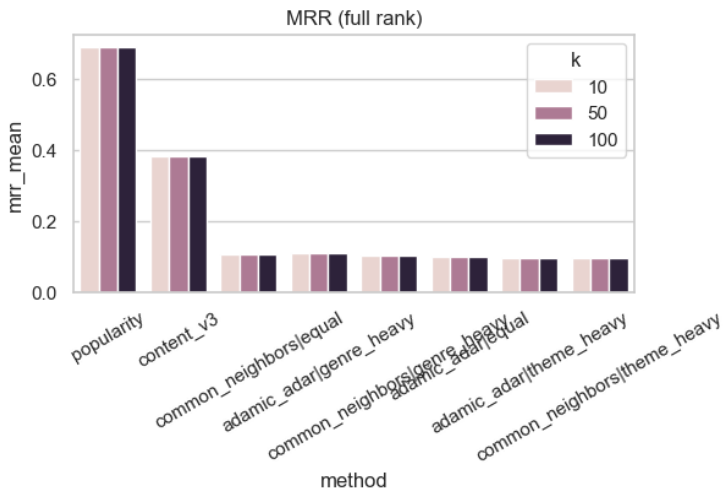
Graph Construction

Node Types:
• Users • Manga • Genres • Themes • Authors

Edge Types:
• User -> Rated -> Manga • User -> Plans to read -> Manga • User -> avoid -> Manga • Manga -> has_genre -> genre • Manga -> has_theme -> theme • Manga -> written_by -> author







Experimental Results

- Popularity baseline performs strongly
- Content_v3 performs very well
- Graph methods perform worse
- Experiment establishes graph methods

Next plan: Stage 3

Graph embeddings over the user-item graph

Methods such as DeepWalk, Node2Vec, Graph Neural Networks

Additions:

- Integrate large, external user dataset
- Maintaing evaluation metrics for fair comparison
- Expand current user database

Summary

Progress:

- Stage 1 baseline system established
- Stage 2 graph heuristics implemented
- Experimental evaluation completed

Key: Simple neighborhood overlap is insufficient for this dataset

Next: Graph representation learning methods

Goal: Improve recommendation quality by learning richer representations